VisionRoot AI Engineer Trainee Program

Day 12 – Daily Task Submission

Date: 12 October 2026
Topic: AI Evaluation
Task Status: Partially Completed (3 of 10 Test Cases Evaluated)

1. What I Learned

Today, I learned the fundamental concepts of AI evaluation and how to assess the quality and reliability of AI-generated responses.

I studied several evaluation metrics, including:

- **Accuracy:** Measures whether the AI response is correct.
- **Relevance:** Measures whether the response addresses the user's question.
- **Groundedness:** Checks whether the response is supported by available information.
- **Faithfulness:** Evaluates whether the AI accurately represents the provided context.
- **Hallucination:** Occurs when an AI generates unsupported or fabricated information.
- **Latency:** Measures the time taken to generate a response.
- **Cost:** Refers to the resources or expenses involved in generating responses.

I also learned how to use a scoring system to evaluate AI responses and identify limitations.

2. What I Implemented

I worked on a Python-based evaluation process for the FastAPI AI application developed during Day 11.

The practical activities included:

1. Preparing an evaluation script using Python.
2. Testing the existing AI application through FastAPI Swagger UI.
3. Sending evaluation questions to the AI API.

4. Collecting actual AI-generated responses.
5. Comparing actual responses with expected answers.
6. Assigning evaluation scores for accuracy, relevance, groundedness, and faithfulness.
7. Recording observations and possible response limitations.

I completed the evaluation of three test questions.

3. Evaluation Results

Test Case 1: Hotel Check-in Time

Question: What is the check-in time?

AI Response: The AI explained that it did not have access to specific booking or hotel information and requested clarification.

Recorded Scores:

- Accuracy: 0/2
- Relevance: 1/2
- Groundedness: 2/2 (Provisional)
- Faithfulness: 2/2 (Provisional)

Observation: The AI did not provide a specific check-in time because the required hotel information was unavailable.

Test Case 2: Hotel Check-out Time

Question: What is the check-out time?

AI Response: The AI suggested a standard check-out time of 11:00 AM and mentioned that times may vary depending on the property.

Recorded Scores:

- Accuracy: 1/2
- Relevance: 2/2
- Groundedness: 1/2 (Provisional)

- Faithfulness: 1/2 (Provisional)

Observation: The response was relevant, but the stated time could not be verified against a specific hotel's policy.

Test Case 3: Deluxe Room Price

Question: What is the price of a Deluxe room?

AI Response: The AI stated that it lacked sufficient information and requested the hotel or property name.

Suggested Scores:

- Accuracy: 1/2
- Relevance: 2/2
- Groundedness: 2/2 (Provisional)
- Faithfulness: 2/2 (Provisional)

Observation: The AI avoided inventing a room price and requested additional information. The suggested scores are not confirmed as saved evaluation results.

Evaluation Limitation: Groundedness and faithfulness were not independently verified against hotel source documents. Therefore, the provisional scores should not be treated as validated evaluation measurements.

4. What Worked

- The FastAPI application successfully processed the tested questions.
- The AI generated responses through the existing API.
- The evaluation process helped identify differences between expected and actual responses.
- The AI requested clarification when specific information was unavailable in two test cases.
- The practical exercise demonstrated how to identify potentially unsupported AI responses.

5. What Failed / Challenges Faced

The main challenge was the lack of hotel-specific information within the evaluation context.

For example, the AI could not provide an exact Deluxe room price or check-in time without knowing the relevant property.

Another limitation was that some responses could not be fully evaluated for groundedness and faithfulness because verified source information was unavailable.

Only three of the planned ten test cases were completed during this session.

6. How I Addressed the Challenges

I reviewed the AI-generated responses and compared them with the intended answers.

I identified situations in which the AI needed additional information and recorded these observations.

I also recognized the importance of creating evaluation questions that match the AI application's actual capabilities and available knowledge.

No prompt modifications or confirmed before-and-after improvements were completed during this session.

7. What I Would Improve

In the next stage, I plan to:

- Complete the remaining seven evaluation questions.
- Prepare a more suitable evaluation dataset with verifiable expected answers.
- Improve the AI instructions to reduce unsupported responses.
- Repeat the evaluation after making improvements.
- Compare results before and after the changes.
- Automate evaluation and response-time measurements where possible.

8. Before-and-After Evaluation

Initial Evaluation: Three test cases reviewed.

Improved Evaluation: Not yet conducted.

Comparison: Not available because the prompt improvement and second evaluation have not been completed.

10. Conclusion

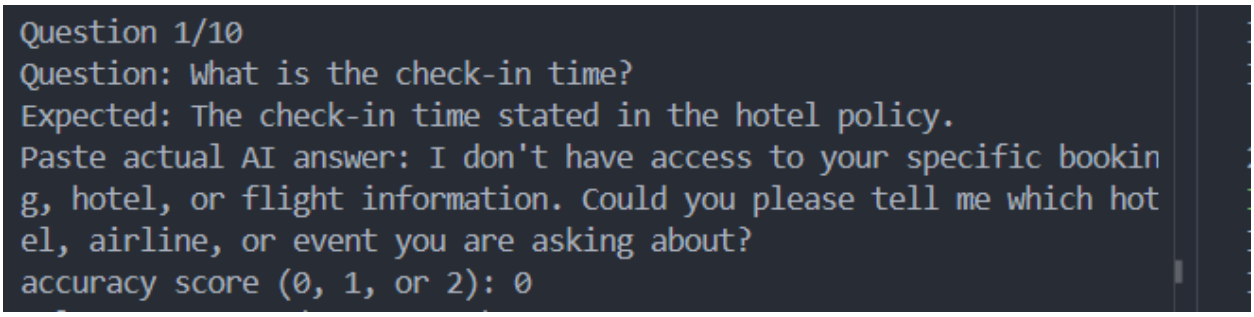
Day 12 provided practical experience in evaluating AI-generated responses using a Python-based workflow and a FastAPI application.

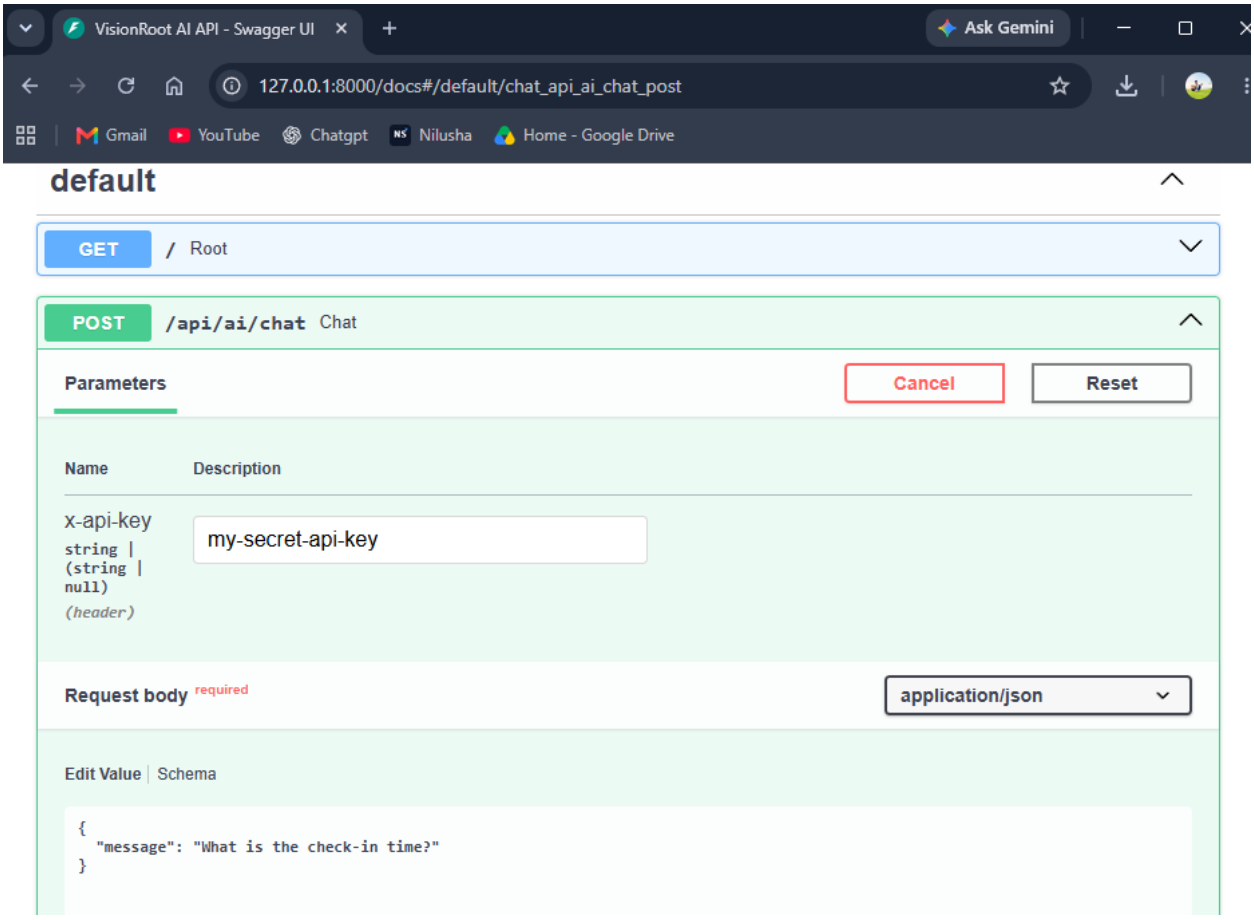
I learned how to assess response quality, recognize missing information, identify potential hallucinations, and document evaluation findings.

Three initial test cases were evaluated, providing a foundation for completing the full evaluation experiment and improving the AI application's reliability.

Overall Status: Partially completed; further testing and comparison are required.

Test 1

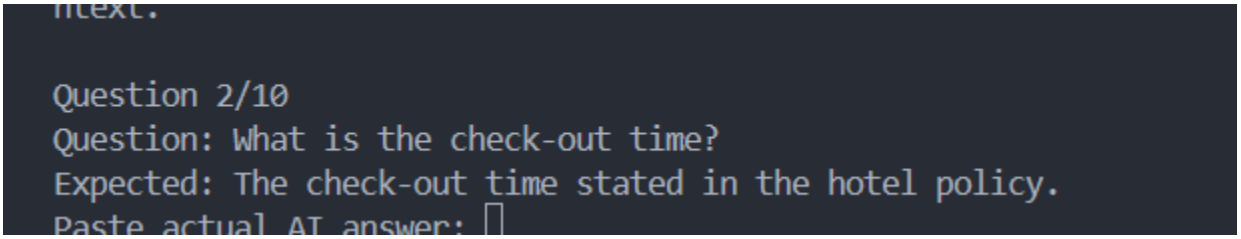


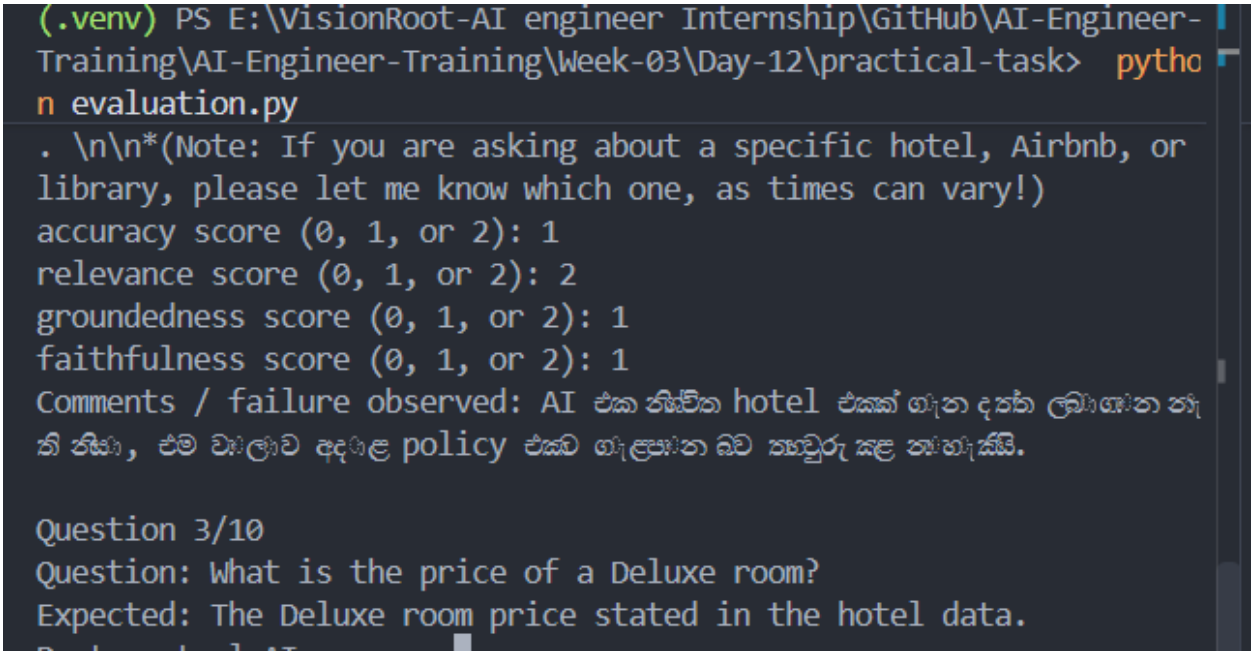



```
n evaluation.py
Expected: The check-in time stated in the hotel policy.
Paste actual AI answer: I don't have access to your specific bookin
g, hotel, or flight information. Could you please tell me which hot
el, airline, or event you are asking about?
accuracy score (0, 1, or 2): 0
relevance score (0, 1, or 2): 1
groundedness score (0, 1, or 2): 2
faithfulness score (0, 1, or 2): 2
Comments / failure observed: The AI could not provide the check-in
time because hotel information was not available in the response co
ntext.

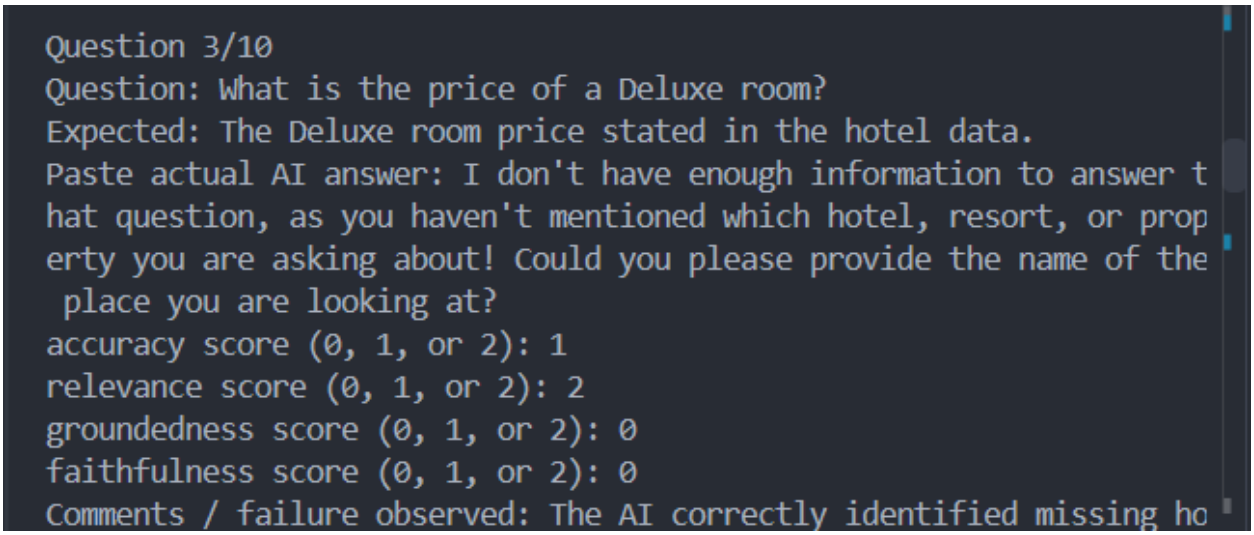
Question 2/10
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Test 2


ntext.
Question 2/10
Question: What is the check-out time?
Expected: The check-out time stated in the hotel policy.
Paste actual AI answer:


(.venv) PS E:\VisionRoot-AI engineer Internship\GitHub\AI-Engineer-Training\AI-Engineer-Training\Week-03\Day-12\practical-task> python evaluation.py
. \n\n*(Note: If you are asking about a specific hotel, Airbnb, or library, please let me know which one, as times can vary!)
accuracy score (0, 1, or 2): 1
relevance score (0, 1, or 2): 2
groundedness score (0, 1, or 2): 1
faithfulness score (0, 1, or 2): 1
Comments / failure observed: AI එක නිශ්චිත hotel එකක් ගැන දත්ත ලබාගන්න නැති නිසා, එම වාලාව අදාළ policy එකට ගැළපෙන බව තහවුරු කළ නොහැකියි.
Question 3/10
Question: What is the price of a Deluxe room?
Expected: The Deluxe room price stated in the hotel data.

Test 3


Question 3/10
Question: What is the price of a Deluxe room?
Expected: The Deluxe room price stated in the hotel data.
Paste actual AI answer: I don't have enough information to answer that question, as you haven't mentioned which hotel, resort, or property you are asking about! Could you please provide the name of the place you are looking at?
accuracy score (0, 1, or 2): 1
relevance score (0, 1, or 2): 2
groundedness score (0, 1, or 2): 0
faithfulness score (0, 1, or 2): 0
Comments / failure observed: The AI correctly identified missing ho